Strategy & module control

A1–A6 • 6 prompts • Suggested time: 9 min

Name: _____ Date: _____ Time: _____

Write observable actions, not general advice. Each response should say what you would do.

A1 · Adaptive route

A student has 12 minutes left with 9 questions still unseen in Module 1. Explain why protecting completion can matter beyond those nine questions.

Action / evidence: Solve visible paths among the 9 unseen items; answer, flag, and move if stalled.

Final answer: Protect reachable correct answers; Module 1 performance also informs Module 2 routing.

A2 · Progress cue

Write one observable sign that you are still making progress after 75 seconds, and one sign that you are only repeating the same attempt.

Action / evidence: Progress: I corrected a sign and have one substitution left.
Repetition: I expanded the same expression twice with no new information.

Final answer: Progress gives a new usable result; repetition does not.

A3 · Provisional response

You flag a multiple-choice item after eliminating two choices. What should you do before leaving the screen, and why?

Action / evidence: A selected response can earn credit if correct; leaving it blank cannot.

Final answer: Select the best remaining choice, confirm it is selected, then flag and move.

A4 · Midpoint check

At question 11, the module clock shows 16:40 remaining. Describe the two facts you would inspect before deciding whether your pace needs adjustment.

Action / evidence: Count unseen/unfinished items; inspect earlier blanks and the time needed to fix flags.

Final answer: Check remaining first-pass workload and the repair cost of earlier blanks/flags.

A5 · Final sweep

Write a final-90-second review order for these categories: unanswered, flagged with no work, and flagged with a nearly complete solution.

Action / evidence: Give every blank a response first; prioritize short, reliable fixes over fresh starts.

Final answer: Unanswered → nearly complete flags → no-work flags → final blank/entry check.

A6 · Entry script

Create a five-second script for a student-produced response that prevents units, variable names, or an entire ordered pair from entering the answer field.

Action / evidence: Keep units, variable names, and ordered pairs on scratch paper; enter one number.

Final answer: “Ask? One number. Sign? Fraction or decimal. No labels. Check.”

Precision warm-up

B1–B6 • 6 questions • Suggested time: 10 min

Name: _____ Date: _____ Time: _____

Identify the requested quantity; show one method line and only then enter the answer.

B1

The endpoints of a segment are (−3, 7) and (9, −1). What is the y-coordinate of the midpoint?

Work: Midpoint y = (7 + (−1))/2 = 6/2 = 3.

Final answer: 3

B2

For $5x^2 + 4x - 12 = 0$, which expression equals the product of the solutions?

- ◯ A. −12/5
- ◯ B. −4/5
- ◯ C. 4/5
- ◯ D. 12/5

Final answer: A · −12/5

Work: Product = c/a = −12/5.

B3

The graph of $y = x^2 - 10x + 17$ has a minimum. What is the minimum value of y?

Work: $y = (x - 5)^2 - 8$; the square is at least 0, so the minimum y is −8.

Final answer: −8

B4

The equation $(x + 3)(x - 8) = 14$ has two real solutions. What is the sum of the solutions?

Work: $x^2 - 5x - 38 = 0$; sum = −b/a = −(−5)/1 = 5.

Final answer: 5

B5

A solution is x = 7/3. Which student-produced response is acceptable?

- ◯ A. x = 7/3
- ◯ B. 7/3 units
- ◯ C. 7/3
- ◯ D. (7, 3)

Final answer: C · 7/3

Work: Enter the number only; no labels or ordered pair.

B6

A Desmos graph shows intersections of $y = 2x^2 - 3$ and $y = 5x + 9$. Which coordinate of each intersection solves the original equation $2x^2 - 3 = 5x + 9$?

- ◯ A. The x-coordinate
- ◯ B. The y-coordinate
- ◯ C. The sum x + y
- ◯ D. The distance from the origin

Final answer: A · The x-coordinate

Work: At each intersection, the same x makes both outputs equal.

Core accuracy | Part 1

C1–C6 • First half of the 24-minute core set

Name: _____ Date: _____ Time: _____

Pass 1: solve visible paths. Pass 2: change representation on flagged questions.

C1

For the linear function f, f(2) = 11 and f(6) = 27. What is f(9)?

Work: Slope = (27 − 11)/(6 − 2) = 4; f(9) = 27 + 4(9 − 6) = 39.

Final answer: 39

C2

A streaming service increases a monthly fee of $15 by 12%. What is the new monthly fee, in dollars?

Work: New fee = 15(1 + 0.12) = 15(1.12) = 16.80.

Final answer: 16.8

C3

The system 3x + 2y = 18 and x − 2y = 6 has solution (x, y). What is the value of x?

Work: Add the equations: 4x = 24, so x = 6.

Final answer: 6

C4

A culture begins with 240 cells and doubles every 3 hours. Which expression gives the number of cells after t hours?

- ◯ A. 240(2)^(t/3)
- ◯ B. 240(3)^(t/2)
- ◯ C. 2(240)^(t/3)
- ◯ D. 240 + 2t/3

Final answer: A · 240(2)^(t/3)

Work: t/3 doubling periods; start with 240.

C5

The function g(x) = (x − 4)² − 9 is graphed in the xy-plane. At what x -value does g attain its minimum?

Work: Vertex = (4, −9); (x − 4)² is smallest when x = 4.

Final answer: 4

C6

One solution of 2x² − 13x + 15 = 0 is x = 5. What is the other solution?

Work: Product of roots = 15/2; other root = (15/2)/5 = 3/2.

Final answer: 3/2

Core accuracy | Part 2

C7–C12 • Second half of the 24-minute core set

Name: _____ Date: _____ Time: _____

Complete C1–C12 as one 24-minute set. Verify what each question actually asks for.

C7

A circle has equation $(x + 2)^2 + (y - 5)^2 = 49$. What is its diameter?

Work: $r^2 = 49 \rightarrow r = 7$; diameter $= 2r = 14$.

Final answer: 14

C8

Paint is mixed in a ratio of 3 parts blue to 5 parts white. If the mixture contains 32 cups total, how many cups are blue?

Work: Total ratio parts $= 3 + 5 = 8$; blue $= 32(3/8) = 12$.

Final answer: 12

C9

The mean of 6, 10, 14, 18, and k is 13. What is k?

Work: Total $= 5(13) = 65$; $k = 65 - (6 + 10 + 14 + 18) = 17$.

Final answer: 17

C10

For $h(x) = 3x^2 + bx - 8$, $h(2) = 10$. What is the value of b?

Work: $3(2)^2 + 2b - 8 = 10 \rightarrow 4 + 2b = 10 \rightarrow b = 3$.

Final answer: 3

C11

The graphs of $y = x^2 - 6x + 4$ and $y = 2x - 8$ intersect at two points. What is the sum of the x -coordinates of the intersection points?

Work: Set outputs equal: $x^2 - 8x + 12 = 0$; sum $= -(-8)/1 = 8$.

Final answer: 8

C12

A rectangle has area 96 square units. Its length is 4 units greater than its width. What is the rectangle’s perimeter?

Work: $w(w + 4) = 96$; width $= 8$, length $= 12$; $P = 2(8 + 12) = 40$.

Final answer: 40

Transfer & tool choice | Part 1

D1–D4 • First half of the 18-minute transfer set

Name: _____ Date: _____ Time: _____

Name the most efficient first representation before solving.

D1 · Constant rate

A tank contains 68 liters of water. It drains at a constant rate of 4.5 liters per minute. Which equation gives the liters L remaining after m minutes?

- ◯ A. L = 68 − 4.5m
- ◯ B. L = 4.5 − 68m
- ◯ C. L = 68(4.5)^m
- ◯ D. L = 68 + 4.5m

First representation: Equation: start + rate × time.

Initial amount = 68; draining rate = −4.5, so L = 68 − 4.5m.

Final answer: A · L = 68 − 4.5m

D2 · Equivalent forms

Which expression is equivalent to $6x^2 - 54$?

- ◯ A. 6(x − 3)(x + 3)
- ◯ B. 6(x − 9)(x + 1)
- ◯ C. 3(2x − 9)(x + 6)
- ◯ D. $6(x - 3)^2$

First representation: Factored form: GCF, then difference of squares.

$6x^2 - 54 = 6(x^2 - 9) = 6(x - 3)(x + 3)$.

Final answer: A · 6(x − 3)(x + 3)

D3 · Similarity

Two similar triangles have corresponding side lengths 8 and 14. If the smaller triangle's area is 40, what is the larger triangle's area?

First representation: Ratio equation: area factor = (side factor)².

Side factor = 14/8 = 7/4; area = $40(7/4)^2 = 245/2 = 122.5$.

Final answer: 122.5 (or 245/2)

D4 · Data interpretation

A data set has median 22. Every value is increased by 7. What is the median of the new data set?

First representation: Shift rule for ordered data.

Adding 7 preserves the order and shifts the middle value(s) by 7.

New median = 22 + 7 = 29.

Final answer: 29

Transfer & tool choice | Part 2

D5–D8 • Second half of the 18-minute transfer set

Name: Date: Time:

Complete D1–D8 as one 18-minute set. Match the graph feature to the ask.

D5 · Parameter

The equation $x^2 + kx + 25 = 0$ has exactly one real solution. What is one possible value of k?

First representation: Discriminant = 0.

$k^2 - 4(1)(25) = 0 \rightarrow k^2 = 100 \rightarrow k = 10$ or -10.

Choose one: $k = 10$ gives $(x + 5)^2 = 0$.

Final answer: 10

D6 · Modeling error

A student solves $(x - 1)(x + 6) = 20$ by graphing only $y = (x - 1)(x + 6)$ and reading its x-intercepts. Diagnose the error and specify the graph feature that should be read instead.

First representation: Two-graph equality model: parabola and horizontal target line.

The original x-intercepts solve $(x - 1)(x + 6) = 0$, not 20.

Add $y = 20$ and read the x-coordinate of each intersection.

Final answer: Wrong target: 0 instead of 20. Read x-values where the parabola meets y = 20.

D7 · Requested quantity

A segment has endpoints (a, 12) and (10, −4). The midpoint's x-coordinate is 7. What is a?

First representation: One-coordinate midpoint equation.

$(a + 10)/2 = 7 \rightarrow a + 10 = 14 \rightarrow a = 4$.

Final answer: 4

D8 · Method comparison

For $4x^2 + 11x - 3 = 0$, compare two methods for finding the sum of the solutions. Which method is fastest, and what calculation does it require?

First representation: Coefficient structure in standard form.

Method 1: sum $= -b/a = -11/4$.

Method 2: $(4x - 1)(x + 3) = 0$; roots 1/4 and −3 sum to $-11/4$.

The coefficient method needs only one signed division.

Final answer: Fastest: coefficient sum, $-b/a = -11/4 = -2.75$.

Bluebook rehearsal & review

E1–E4 • 4 decisions • Suggested time: 7 min

Name: _____ Date: _____ Time: _____

Decide, act, and state the evidence that would justify your action.

E1 · Flag or finish

You have a correct equation but made an arithmetic slip, and 40 seconds have elapsed. Should you repair now or flag and move? Defend the choice using expected time to completion.

Decision / reason: If the slip is located, a 10–20-second repair may be worth finishing now. If finding the error will take uncertain time, enter a provisional response and flag.

Final answer: Repair if the finish is brief and time permits; otherwise flag and move.

E2 · Graph window

A quadratic appears to have no x-intercepts in the default Desmos window. List two checks you must make before concluding that the equation has no real solutions.

Decision / reason: Verify signs, coefficients, parentheses, and the correct zero/intersection target. Zoom or pan to the vertex and possible roots; use the discriminant for certainty.

Final answer: Check (1) the entered equation/target and (2) the relevant graph window.

E3 · Review page

The review screen shows one unanswered item and three flagged items with responses. You have 55 seconds. Write the exact order of operations.

Decision / reason: Protect the blank first, then use the time on the shortest promising repair. Example budget: 10 seconds for the blank, 35 for a repair, 10 to verify.

Final answer: Fill blank → quickest promising flag → other flags if time → verify entries.

E4 · SPR precision

Your work gives the point (−2, 11), but the prompt asks for the maximum value of the function. What belongs in the response box, and what should stay on scratch paper?

Decision / reason: The maximum value is the y-coordinate, 11; −2 is where it occurs.

Final answer: Enter 11; leave (−2, 11) and x = −2 on scratch paper.

PROCESS DEBRIEF

Time used: _____ Pace changed at: _____ Flagged IDs: _____

On Pass 2, I switched from _____ to _____.

ERROR LOG / Correct up to three questions after checking your work

ID	First wrong decision	Type	Next-time cue